

Policy frameworks for district heating: A comprehensive overview and analysis of regulations and support measures across Europe

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ABSTRACT

This paper presents an overview of policy frameworks for district heating (DH) in Europe. We develop a classification for policy aspects addressing DH, comprising regulation of ownership, prices, metering, consumer grid connection, third party access as well as support schemes and carbon taxes. This classification builds on existing literature and expert assessments collected in an online survey and interviews. The relevance of the paper lies in giving a comprehensive picture of the existing policy frameworks for DH in 23 European countries. For the overview, the countries are clustered along assigned categories. It is shown that the policy frameworks for DH can be clustered into five distinguishable groups. While some countries apply very strict regulations in special DH laws, others rely on less regulatory intervention, both with varying degrees of support for DH in place. The different policy frameworks are discussed in the context of the diffusion of DH and the integration of renewable and waste heat in DH. This analysis shows that high shares of DH and high shares of renewable and waste heat in DH can be observed in countries with a high degree of regulation as well as in countries with less regulatory intervention in DH markets.

1. Introduction

The heating and cooling sector accounts for about 50% of the total European final energy consumption (Heat Roadmap Europe, 2017). Hence, the decarbonisation of heating and cooling is crucial for achieving net-zero greenhouse gas emissions by 2050. District heating (DH) networks will play a decisive role in the decarbonisation of the heating sector by enabling the integration of renewable and waste heat on a larger and more centralized scale than decentralised decarbonisation solutions, which have their limitations, especially in urban areas (Bürger et al., 2019; Connolly et al., 2014). Hence, the European Commission envisages the transformation of DH and obliges the Member States (MS) to increase the share of renewable and waste heat in DH systems or enable third party renewable or waste heat generators to access DH (Renewable Energy Directive, 2018/2001, RED II). However,

in contrast to the gas and electricity markets, no European market standardisation and liberalisation has been carried out for DH.

Many technical analyses of DH are available, including approaches looking at individual networks as well as analyses focusing on description and categorization of DH (i.e. 4th or 5th generation DH) (e.g. Averfalk and Werner, 2017; Buffa et al., 2019; Li and Nord, 2018; Lund et al., 2021; Lund et al., 2018; Lund et al., 2014; Rhein et al., 2019). However, only a few scientific publications focus on policy or regulatory frameworks for DH. Wissner (2014) examines the possibility of regulating DH systems and the difficulties associated with practical implementation. The article approaches the question of price and third party access (TPA) regulations from a theoretical microeconomic perspective and matches the results with the reality of the DH sector in Germany. Furthermore, Werner (2017a) outlines the background and current status of DH in Sweden as well as the position of DH worldwide (2017b),

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with some in-depth insights into conditions in Europe. Both papers provide broad reviews of DH, but only limited insights into regulatory issues. In addition, specific regulatory topics are discussed more intensively in some papers. Bürger et al. (2019) and Holzleitner et al. (2020), for example, develop a better understanding of the complexity associated with the introduction of TPA. Moreover, some papers focus on specific regions; Büchele et al. (2018), for example, model different policy scenarios for Brasov, Romania. In addition to scientific publications, there are some project reports such as Andrews et al. (2012), Steinbach et al. (2017), Chassein et al. (2017), or Bacquet et al. (2022b), which sketch the current status of DH, and provide background information on potentials, barriers, best practices, state of the art and policy measures. The reports show that the current political frameworks of DH in many European countries still offer few incentives to address existing challenges. Overall, in the existing literature, little attention is paid to a comprehensive policy framework addressing barriers and enabling large-scale implementation of technically and economically efficient DH networks for sustainable heat supply.

This paper fills the research gap by providing a comprehensive, in-depth overview of applied regulations for DH in Europe. In addition to the aspects discussed in the literature so far, further regulatory aspects addressing DH are identified and analysed. This paper looks at the regulation of ownership, prices, metering, consumer grid connection, TPA as well as support schemes and carbon taxes. For each aspect, we elaborate a classification, displaying different categories of the regulatory aspects, for example, no legislation in place, some form of regulation or a strict regulatory approach. Based on the classification, we provide an overview of the applied regulation and support schemes, focusing on issues that are commonly discussed in literature and where data could be collected. Moreover, we discuss the policy frameworks in the context of the integration of renewable and waste heat in DH and the diffusion of DH. With this discussion, we try to determine whether specific regulatory design elements coincide with a low or high share of renewable and waste heat in DH and a low or high share of citizens served by DH. So far, DH data is lacking, i.e. an overview of (historic) regulations as well as (historic) DH shares is missing. This data background underlines the high relevance of our analysis. Thus, this paper is among the first papers, which provides a comprehensive overview of policy frameworks for DH and analyses best practices.

The remainder of the paper is structured as follows: Section 2 provides a description of the data collection process and the methodology used in this paper. In section 3, a classification for the policy aspects addressing DH is derived. Section 4 presents and discusses an overview of the policies based on the classification developed. Similar policy frameworks are indicated and their impact is discussed. Section 5 provides conclusions and policy implications.

2. Data collection and methodology

The paper approaches the question of regulation and support of DH in the MS, Iceland, Norway and the UK. Countries with no or a low share of DH, i.e. Greece, Portugal, Ireland, Malta and Cyprus or countries with only little available information, i.e. Belgium and Luxembourg are excluded. All in all, this paper analyses regulations and support schemes for DH of 23 countries.

In this paper, we use a mixed-method approach. First, relevant regulatory issues were selected. For each issue, an in-depth literature review was conducted and gaps were filled with an extensive online survey and interviews with national DH experts. The timeframe of the data collection was from summer 2019 to early 2021. The data collection was done in the framework of the European project 'DHC Trend' (see acknowledgements and Bacquet et al., 2022b). The collected information was then used in a morphological analysis to develop a classification of DH policies. Finally, a clustering approach, using the output of the morphological analysis as input data, was applied with the aim to identify similar policy frameworks for DH. Lastly, the identified policy

frameworks were discussed in the framework of selected indicators. In the following, each method is described in more detail.

2.1. Selection of regulatory issues

We analyse the following regulatory issues: (1) ownership and operation of DH networks, (2) heat sale prices for DH consumers, (3) metering of heat consumption (4) consumer grid connection and usage, (5) TPA as well as (6) support measures for DH and national carbon taxes. This selection is based on the rationale that these are elementary aspects that determine in most cases the policy framework for DH. The question of ownership based on current implementations and economic theory is highly relevant, i.e. who is allowed to own and operate a network and what requirements need to be fulfilled. Regulation of prices and TPA are also discussed in the literature in the context of natural monopolies (see e.g. Wissner, 2014). Strongly connected with the regulation of prices is the regulation of metering of consumed heat for billing purposes. The issue of consumer rights and especially consumer grid connection has, to our knowledge, not been discussed in the scientific literature, but is often addressed in the regulatory framework of DH and can have a huge impact on the DH market. Lastly, support measures, like investment support schemes for DH infrastructure or indirect support for renewables in DH through carbon taxes have an impact on the success of decarbonising DH. Other aspects influencing DH such as spatial planning or building regulations are not discussed.

2.2. Literature review

The literature review is based on existing scientific papers (e.g. Bürger et al., 2019; Sarma and Bazbauers, 2016; Söderholm and Wårell, 2011; Wissner, 2014), studies and reports (e.g. Castellazzi, 2017; Euroheat & Power, 2019; Patronen et al., 2017), MS plans (progress reports, national energy and climate plans) and databases (RES Legal¹ and IEA policy database²). In addition, national acts and legal documents, national reports and websites from energy agencies, authorities and national associations or companies were included. These country-specific sources can be found in the country factsheets in Bacquet et al. (2022a) (project report 'DHC Trend'; The data collection took place in the framework of this project; see acknowledgements).

Key words for the literature search were, for example, 'district heating regulation', 'district heating price regulation', 'regulation of heat metering' as well as phrases such as 'mandatory grid connection to district heating in Germany'. Documents in non-English languages were translated with machine translation software. The collected information from the literature was verified and complemented with an online survey and interviews.

2.3. Online survey and interviews

The online survey included 20 questions on the current legislation affecting DH in the respective country, whereby both closed multiple-choice and open questions were used. In some cases, additional semi-structured interviews were conducted to better understand the legislation. In total, 50 experts from regulatory authorities, energy agencies, district heating associations or operators answered the questions of the online survey or participated in an interview. The organisations that participated in the survey or an interview are listed in appendix B. The country factsheets in Bacquet et al. (2022a) show which information was provided through the survey or interviews.

¹ <http://www.res-legal.eu>.

² <https://www.iea.org/policies>.

2.4. Morphological analysis

For each regulatory issue, a classification of different forms or intensities of regulations and support schemes is presented. A morphological analysis is used for this classification, orientated on the approaches in Zwicky and Wilson (1967), Lay et al. (2009) and Kley et al. (2011). In this method, a table is laid down and parameters are defined. Afterwards, for each parameter, the values for possible configurations, i.e. elements that belong to the parameters, are added. In this paper, the elements represent a form of regulation or support. Each element is connected to at least one observation, i.e. in at least one country of the 23 countries the form of regulation or support exists. The output of the analysis are so-called morphological boxes. The morphological boxes should be interpreted as selection matrixes, i.e. different combinations of the elements can be selected.

2.5. Clustering analysis

Based on the morphological analysis, we present an overview of existing regulations and support schemes. Therefore, the first parameter of each morphological box is selected and translated into an ordinal scale with a 1-of-3 encoding of the elements. The ordinal dataset is then grouped with a clustering algorithm to identify similar policy frameworks between the countries.

Ideally, all parameters of the morphological boxes should be used in the clustering, but this is not possible due to a lack of data. Therefore, we used parameters that are commonly discussed in literature and where data was at least partly available. Data gaps were filled with a survey and interviews (see section 2.3). Each country was evaluated based on the legislation in force during the data collection period, i.e. mainly in 2019, and for the selected parameter a value between 1 and 3 was assigned. The resulting data set is shown in appendix C. Even though not all of the parameters of the morphological boxes can be presented, the selected parameters and related elements enable a comprehensive overview of policies addressing DH.

The clustering was performed using an agglomerative clustering algorithm provided by the Python package SciPy (Pedregosa et al., 2011). In agglomerative clustering, the algorithm calculates the dissimilarity between all elements and gradually combines two elements with the least dissimilarity into a cluster. This formed cluster is then used again in the next iteration. Various schemes for the dissimilarity calculation and linkage types are available (see e.g. Ward, 1963). We used Ward's minimum variance method and the Euclidean distance. Further details on the clustering analysis, including a sensitivity analysis with different clusterings, reasoning for the selected algorithm, are presented in appendix C.

2.6. Discussion of the impact of policies

Finally, the results of the clustering, i.e. country groups with different policy frameworks, are discussed in the context of indicators representing the success of the policy framework. As indicators, we use the share of renewable and waste heat in DH and the share of citizens served by DH. It is important to note that this analysis is limited as only the status quo of policies and shares could be considered due to a lack of data. However, historical developments (and other factors) are taken into account in the discussion. Despite the limitations, this paper assesses the different policy frameworks and identifies best practices for DH.

3. Policy frameworks for district heating

To assess the suitability of different policies, it is necessary to have a structured overview of possible design elements for each policy aspect. Hence, in this section we describe and categorise design elements in order to develop a classification for the different forms of regulations

and support measures affecting DH. The categories are based on existing literature and on expert knowledge collected in a survey and interviews (see section 2).

3.1. Ownership and operation of networks

So far, there is only little research on the regulation of ownership and operation of DH networks. Egüez (2021) and Biggar and Söderberg (2020) analyse correlations between ownership and prices, focusing on the Swedish DH market. Bacquet et al. (2022b) show that in most European countries DH networks are operated by vertically integrated companies, owned by municipalities or by private companies in which municipalities hold a major share. However, in some countries (e.g. Poland, Slovakia or Spain) a mix between public and private ownership is common, whereby large networks are often owned and operated by large energy companies, whilst smaller networks with small plants are typically owned by municipalities (Bacquet et al., 2022b).

At the same time, in the majority of the countries, DH systems need some sort of authorisation, e.g. a licence, permit or concession (Bacquet et al., 2022b). In some countries, the required licence is connected to the size or ownership structure of the DH network. This is for example the case in Norway, where a license is required for DH with a total power output of more than 10 MW under the Norwegian Energy Act. The authorisation process also varies: In Denmark, operators of new DH networks can submit a project proposal to the municipality, and approval of the proposal is in accordance with the Danish Heat Supply Act and automatically leads to a licence to supply heat. Similarly, to operate a public DH network in France, a tender procedure must be organized by the local authority and the awarded operator gets a contractual right to operate the DH network (Bacquet et al., 2022b). The development of private networks is also possible, whereby operators do not need any particular approval other than the regular permit for the use of public domain, where applicable (Bacquet et al., 2022b). Moreover, there are a number of countries, like Estonia, Hungary, Lithuania, and Poland, where all DH networks must have an operating licence, which is usually granted according to a standardised procedure. Furthermore, in some countries, DH networks need only to be registered. This is the case for example in Germany, where companies, including DH companies, need to be registered in the German Commercial Register. In addition to the company registration, a permit must be obtained in general for (large) plants under the German Immission Control Act. On the other side of the spectrum, there are only a few countries that do not require any forms of licencing or registering (e.g. Finland).

Fig. 1 shows the developed classification for regulation of ownership and operation of DH in a morphological box. As described in section 2, this morphological box should be interpreted as a selection matrix, i.e. different combinations of the diverse elements are possible. A combination could be, for example, that large networks require authorisation for construction and operation following a standardised procedure at the time of initial operation. In section 4, the elements of the first parameter of the morphological box are used in the clustering to display which regulation is most common.

3.2. Prices for consumers

DH prices differ according to the network and possibly even according to the consumers. An initial estimate of average national DH prices in Europe has been provided by Werner (2016). However, in most countries, prices are not collected or published, so there is a lack of DH price data. At the same time, the European regulation on price transparency only covers electricity and gas prices, resulting in a lack of overarching regulations for DH prices.

There are a few scientific contributions that provide insights to DH price regulations (e.g. Sarma and Bazbauers, 2016; Werner, 2017b; Wissner, 2014). Wissner (2014) explores ex-ante and ex-post price regulation. In the case of ex-ante regulation, prices need to be approved

Parameter	Elements			
Main form of regulation*	No specific regulation of ownership	Requirement for registration for operators	Authorization, i.e. license, permit or concession	
Authorisation procedure	No specific procedure defined	Standard forms and processes	Tender organized by authority	
Scope of authorisation	Authorisation for construction of DH grid	Authorisation for construction of generation plants	Authorisation for operation of DH grid	Authorisation for operation of generation plants
Timing of authorisation	Authorisation by initial operation	Authorisation for specific changes, e.g. new plants or pipes	Authorisation for each change	
Size of addressed networks	Only big DH networks or networks with large generation plants	Only small DH networks	All sizes	
Ownership of addressed networks	Only public networks	Only private DH networks	All ownership models	

* Data available and used in section 4. Colour of field reflects intensity of policy, i.e. darker colour corresponds to stricter regulation.

Fig. 1. Morphological box for the regulation of ownership and operation of DH (building on Bacquet et al., 2022b).

by an authority before they are applied. By ex-post regulation, the competent authority checks ex-post whether the prices were reasonable or in compliance with the defined price rules. Ex-post price control is in most cases based on cartel law, i.e. the ability to investigate DH prices in order to protect consumers. This means, that the ex-post control process often only takes place if there are reasonable grounds to suspect that an operator is charging disproportionate prices.

Nevertheless, several countries rely on price regulation going further than cartel law supervision (see section 4). In a few countries, prices are even administratively set by the regulatory authority. Furthermore, several countries defined some form of price rules, i.e. price caps, price

adjustment clauses or rather an open phrase like 'prices must be set in a proportionate and cost-based manner'. In Norway, for example, the DH price cannot be higher than an alternative cost of other heating sources in the respective licensed DH area according to the Energy Act. Germany relies on price adjustment-clauses, i.e. prices orientated on mostly gas and coal prices, defined in the Ordinance on General Terms and Conditions for the Supply of District Heating. Denmark uses a non-profit rule, i.e. prices can only cover necessary costs, but, with a few minor exemptions, not a profit under the Heat Supply Act. Additionally, some countries rely on non-discrimination or transparency rules. Finland, for example, does not have specific legislation concerning the pricing of DH,

Parameter	Elements			
General approach* (who sets the price)	Freely formed prices, i.e. no specific regulation	Price is defined by operator according to regulation	Price is defined or set by regulator	
Form of pricing (price calculation rule)	(Fix) Price cap	Price adjustment clauses with indicators for adjustment (e.g. gas price)	Defined method for calculation, e.g. list of price components or specified model	Non-profit rule
Terms of prices	No terms defined	Transparency and information requirements	Equality regulation, i.e. same price for all consumers	
Price control	Ex-post control, i.e. prices can be controlled even if already applied		Ex-ante control, i.e. prices must be approved before applied (similar to price setting by the regulator)	
Timing of price control	Only on request	Mandatory on regular occasions	Mandatory each time of price change	
Addressed networks	Only specific networks (e.g. large networks or public networks)		All networks	

* Data available and used in section 4. Colour of field reflects intensity of policy, i.e. darker colour corresponds to stricter regulation.

Fig. 2. Morphological box for the regulation of DH prices (building on Bacquet et al., 2022b).

but in line with competition legislation, a dominant market position requires that pricing is the same for all customers.

Existing literature and practical experience show that a distinction can be made between more regulated and liberalised approaches to price regulation, resulting in strictly structured or freely formed prices. From this, we derived several different parameters with various elements, which are shown in Fig. 2. The elements of the first parameter of this classification are used to illustrate common approaches to price regulation (see section 4).

3.3. Metering of heat consumption

The metering of the heat consumed at the supplier and customer interface for billing purposes is quite highly regulated in most MS, which is mainly a result of European legislation. The European Energy Efficiency Directive (2018/2002, EED), regulates metering and stipulates that end-users must be provided with competitively priced meters that accurately reflect their heat consumption (Art 9a EED). The EED therefore underlines the importance of real metering of consumed heat. In multi-apartment and multi-purpose buildings, individual consumption meters must be installed to measure the consumption for each building unit, where technically feasible and cost effective (Art 9b EED). Moreover, it should be possible for meters and heat cost allocators to be readable remotely by 1 January 2027, if technically feasible and cost effective (Art 9c EED). According to Martins (2017), the implementation of the EED seems to be very difficult, as in many situations several meters or heat cost allocators are necessary to determine the heat consumption of a single end-user.

Looking at national legislation, most countries only refer to the requirements defined in the EED and state that these requirements are applied. Only a few countries are more precise in their legislation. In Bulgaria the Energy Act defines that DH companies must meter the heat consumption every month and provide data to heat cost allocation companies. Lithuania and Estonia have already specified precise national regulations regarding remotely readable meters. Despite the European legislation, regulation of metering is lacking in Greece, Italy, and Spain. In Spain, details on metering may be clarified in public tenders or in private contracts. Norway and Iceland also have not yet introduced legislation regarding metering of DH.

Based on the existing literature and the practical experience, we derived several parameters with various elements (see Fig. 3). The

elements of the first parameter are again used in section 4.

3.4. Consumer grid connection and usage

Grid connection and consumer's usage is a highly relevant topic for DH, as DH systems are often confronted with the challenge that not enough citizens are motivated to connect to the network, leading to low profitability of the systems (Bacquet et al., 2022b). To address this challenge, some countries have introduced mandatory grid connection for consumers, which means that consumers are obliged to connect to their local DH system under certain conditions. In most cases, the obligation applies only to new buildings in certain regions, for example, when a municipality impose some form of DH zoning (Bacquet et al., 2022b).

Mandatory connection for the consumer must be well justified. In France, for example, a mandatory connection and usage can only be implemented when a DH network is classified, which can be done by the local authority, when the system is economically balanced, the energy mix uses more than 50% of renewable or waste heat and all heat deliveries are metered (French Energy Code). New and renovated buildings located within the classified area are obliged to be connected to the DH network and to use DH as a heating source (Bacquet et al., 2022b). Similarly, in Denmark, municipalities were able to impose a mandatory connection to DH in accordance with the Heat Supply Act, when it proves to be the most cost-efficient heating solution. In 2019, this option was removed from the Heat Supply Act, however the existing obligations, which affect around half of all DH customers, remained in place (Euroheat and Power, 2019). In contrast, a municipality in Germany can stipulate by statute that every property within a certain area must be connected, for example, under the precondition of 'common good'. It is estimated that about 35% of the connected customers fall under such regulations and are (partly) obliged to connect and use their local DH system (Schweikardt et al., 2012). In Norway, the municipal council can also decide to order mandatory connections for licensed networks. That being said, Norwegian building owners with mandatory connections are not obliged to use DH as a heating source even if they are connected to the network.

Nevertheless, most countries do not rely on mandatory grid connection of consumers, but have introduced some regulations, e.g. regarding the connection process or delivering contracts. For example, in Bulgaria, there is no mandatory connection for consumers, but the

Parameter	Elements			
Main form of (national) regulation*	No regulation on metering DH	Regulation for heat meters following EU legislation	Regulation going further than EU legislation for smart heat meters and remote control	
Installation	No mandatory installations of meters	Mandatory installation of meters	Mandatory installation of smart heat meters and remote control	
Type of meter	No standards or regulations for the type of meter	Regulation of type of meters, i.e. standards	Regulation of type of smart meters	
Frequency of providing meter data (and billing)	No frequency defined	Yearly	Monthly	Daily
Individualised metering	No regulation on individualised metering	Regulation on individualised metering in a multi-apartment / multi-use building		
Allocation rules	No heat cost allocation rules	Heat cost allocation rules for metering in a multi-apartment / multi-use building		

* Data available and used in section 4. Colour of field reflects intensity of policy, i.e. darker colour corresponds to stricter regulation.

Fig. 3. Morphological box for regulation of metering of DH (building on Bacquet et al., 2022b).

process of connection is well regulated in the Energy Act and detailed in an ordinance.

Fig. 4 shows the developed classification for the regulation of grid connection and usage of consumers of DH. The elements of the first parameter of the classification are again used in section 4 to show and analyse similar regulatory approaches.

3.5. Third party access to networks

Unlike the gas and electricity market, DH markets are not required to be unbundled under EU legislation. Hence, in most countries, DH is an integrated service, i.e. one single company generates heat, operates the local DH grid, and sells heat at the same time. Third parties, i.e. independent producers, have in most cases no access to DH systems.

So far, there has been only little research on TPA to DH networks. Söderholm and Wårell (2011) show that there are many conceptual options to open DH networks for third parties. This includes the option of opening DH to competition on the supply side, referred to as 'full TPA' as well as opening on the generation side, referred to as 'producer TPA'. Moreover, TPA concepts can be subdivided into 'network access model', 'extended producer market' and 'single buyer model' (Bürger et al., 2019; Söderholm and Wårell, 2011). In the network access model, producers have access to networks and they provide the heat they produce to their end customers. In contrast, in the single buyer model, producers are entitled to feed heat into the network, while the grid operator is obliged to accept and pay for the heat. In this model, consumers do not have any choice between different suppliers, they are all supplied by the single buyer, i.e. the grid operator. The 'extended producer market' is a form of the single buyer model, extended by high transparency rules for all market actors. The idea of this model is that clear unbundling rules and high transparency requirements can reduce the regulation efforts.

Furthermore, Bürger et al. (2019) develop a better understanding of the complexity associated with the introduction of TPA and also describe different conceptual models for grid access. They distinguish between negotiated and regulated TPA. Negotiated voluntary network access is a form of voluntary access in which the heat dispatch order to the DH network is determined freely. The term negotiated implies that the DH

network owners are required to negotiate about access to the network with the heat producer. In contrast, negotiated mandatory network access is a clear obligation for DH grid operators to enable grid access, whereby the conditions for grid access are still freely negotiated between the parties involved. Fully regulated network access also includes an obligation for DH grid operators to enable grid access and in addition the access conditions are predefined. If these predefined conditions are met by the heat producer requesting grid access, the operator is obliged to provide the access.

Looking into the praxis, there are first regulations addressing TPA to DH in the European legislation. The RED II includes in Art 24 (4) two options to increase the share of renewable energies in district heating and cooling (DHC) networks: (1) either by the implementation of measures aimed at increasing the share of renewables and from waste heat and cold in DHC by 1% per year, (2) or by granting producers of renewable or waste heat and cold access to the grid (i.e. TPA). So far, around half of the countries introduced some regulation on TPA (Bacquet et al., 2022b). However, Holzleitner et al. (2020) concluded that the option for TPA in the RED II did not fundamentally change the third party generators' situations as operators can still deny a feed-in request. This conclusion is backed by the empirical analysis in Bacquet et al. (2022b). Art 24 (5) of the RED II lists conditions on which basis TPA can be refused, e.g. if technical or economic conditions cannot be met by the supplier. Additionally, Art 26 (6) states, that MS may exempt operators, as long as the networks meet certain criteria (e.g. for efficient DH). Under the 2021 revision of the RED II, TPA for networks with more than 25 MW is currently under discussion (European Commission, 2021).

The existing national regulations regarding TPA are in line with the European legislation. In Latvia, for example, DH operators are obliged to purchase heat from independent producers under the Energy Law. However, this obligation only applies, if the heat price offered is lower than the price of the grid operator, there is enough heating demand in the network and the producer can meet technical requirements. Another example is Lithuania, which also introduced an obligation to connect independent producers, as defined in the Law on the Heat Sector. Access can, however, also be denied, i.e. it is only obligatory if the heat is cheaper than the heat produced by the supplier itself, if it satisfies environmental and quality requirements as well as standards for the

Parameter	Elements			
Regulation of grid connection for consumer*	No specific regulation, i.e. voluntary connection with no further regulations	Voluntary connection, but legislation for the connection process, i.e. technical standards or for connection fees		Mandatory grid connection for consumers (under certain conditions, i.e. zoning)
Regulation of grid usage for consumers	No specific regulation, i.e. voluntary usage and individual contracts	Regulation regarding delivery contracts		Obligation to use DH
(Pre-)condition for mandatory connection and usage	No condition	Only if DH is the most cost-efficient heating solution	Only if the DH network is authorized / classified	Further criteria, i.e. definition of carbon emission level
Regulation for DH operator	No obligations for DH provider defined in legislation		Obligations to connect consumers (e.g. under non-discriminatory conditions)	
Addressed networks (or area)	Only specific networks (e.g. large, public, or classified networks)		All areas / networks	
Addressed buildings	Only renovated and new buildings		All buildings	

* Data available and used in section 4. Colour of field reflects intensity of policy, i.e. darker colour corresponds to stricter regulation.

Fig. 4. Morphological box for regulation of DH consumer grid connection and usage (building on Bacquet et al., 2022b).

Parameter	Elements			
Regulation of grid access for independent producers*	No regulation, i.e. voluntary network access		Regulated TPA with an obligation to enable grid access for renewable or waste heat producers	Regulated TPA with an obligation to enable grid access for all heat generators
(Pre-)conditions for grid access	No conditions	Technical conditions, i.e. security of supply	Economic conditions, i.e. low price	Environmental and quality requirements
Models for grid usage	Single buyer model		Extended producer market	Network access models
Purchase regulations	No priority purchase		Priority purchase for renewable heat	
Contracts	No regulation of contracts		Regulated contracts, i.e. fair conditions in contracts, equal terms	
Addressed networks	Only specific networks (e.g. large or public networks)		All networks	

* Data available and used in section 4. Colour of field reflects intensity of policy, i.e. darker colour corresponds to stricter regulation.

Fig. 5. Morphological box for the regulation of TPA (building on Bacquet et al., 2022b; Bürger et al., 2019; Söderholm and Wårell, 2011).

Parameter	Elements						
Addressed area of DH*	Grid infrastructure		Renewable and efficient generation	Research, technology development, demonstration projects		Connecting end-consumers	
General type of support scheme	Funding / grants	Feed-in tariffs ³	Feed-in premiums	Low-interest loans	Tax exemptions	Taxes on non-RES, i.e. carbon tax*	Quota systems
Addressed costs	CAPEX			OPEX			
Type of payment	Based on generation		Based on capacity		Based on consumption (i.e. taxes)		
Frequency of payment	Once			Regularly			
Determination of the support level	Administrative			Competitive (e.g. auction)			
Technology specification	Multi-technology			Technology specific			
Beneficiaries	DHC operator		DH generation plant		End-consumer		

* Data available and used in section 4. Colour of field does not reflect intensity.

Fig. 6. Morphological box for DH support measures (building on Bacquet et al., 2022b and Winkler et al., 2016).³

security of supply. In contrast, several countries do not have an obligation to connect third party producers, but specific statutory frameworks for grid connection of independent parties. This is the case for example in Croatia, where the grid connection of independent heat producers must be licenced and registered by the authority.

³ In case of a feed-in tariff, plant operators receive a payment for each unit of heat fed into a DH grid. In case of a feed-in premium, plant operators have to market their heat directly and receive an additional payment on top of their market revenues.

Furthermore, in the Netherlands, there is legislation in place, that requires DH operators to negotiate grid access at the very least (Bacquet et al., 2022b).

Combing the existing literature and the practical experience, several parameters with various elements can be distinguished, which illustrate the complexity of TPA (see Fig. 5). The elements of the first parameter of the classification are again used in section 4 for the overview of policy frameworks.

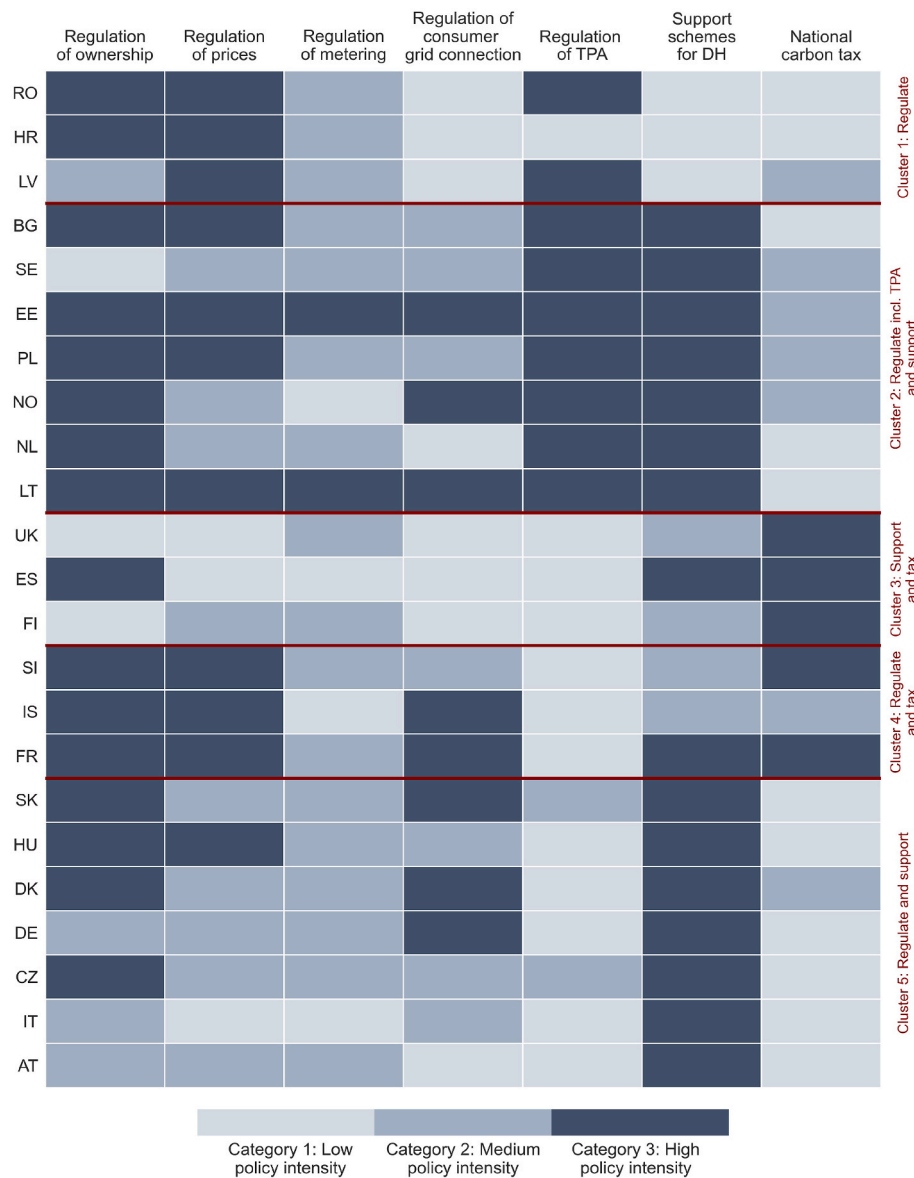


Fig. 7. Overview of applied policy frameworks in the countries ordered by cluster (red lines indicate the division into the different clusters; data partly from [Bacquet et al., 2022b](#)).⁴ (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.6. Support measures and carbon taxes

Support measures for renewables and carbon taxes aim to overcome market failure and incentivize increased investment in renewable technologies. With this overarching goal in mind, several countries implemented support schemes for DH networks based on efficient and

renewable generation. This is laid down in a number of project reports that include analysis on support measures for DH (e.g. [Andrews et al., 2012](#); [Bacquet et al., 2022b](#); [Galindo Fernández et al., 2021](#); [Steinbach et al., 2017](#)). [Bacquet et al. \(2022b\)](#), for example, present in their report the availability of the support schemes and discuss their intensity in terms of quality. They distinguish four types of support schemes for DH networks, comprising financial support (1) for grid infrastructure, (2) for renewable and efficient generation, (3) for research, technology development and demonstration of innovative systems as well as (4) for connecting end-users to networks. Accordingly, most MS have subsidies and financial incentives targeting grid infrastructure as well as for renewable and efficient DH generation ([Bacquet et al., 2022b](#)). The most commonly used types of instruments are financing grants, premiums, low-interest loans or tax exemptions ([Bacquet et al., 2022b](#)). In contrast, financial incentives for research as well as support for the connection of end-users seem to be less common ([Bacquet et al., 2022b](#)). Yet, some countries have implemented comprehensive support schemes, addressing several aspects of DH networks. For example, the German program 'Heating Network Systems 4.0' provides financial support for DH grids with a high share of renewables ([BAFA, 2021](#)). Whereby funding is

⁴ Comments on the country assessment: In Denmark, municipalities were able to impose a compulsory grid access for consumers until 2019. This option has been removed from the Heat Supply Act, but existing obligations, covering around half of all DH customers, remain in force. In France, heat production plants with a capacity over 30 kW located within priority development areas are obliged to be connected to the heating network. However, as this regulation refers only to priority development areas, which usually do not include potential third party supplier, TPA is in practice not regulated. In Spain, the prices for DH are locally regulated if the network is developed under a concession regime. In the case of private networks, prices are not regulated. In Norway, only DH networks with a capacity of more than 10 MW are required to hold a concession and only networks with a concession can apply mandatory grid access for consumers.

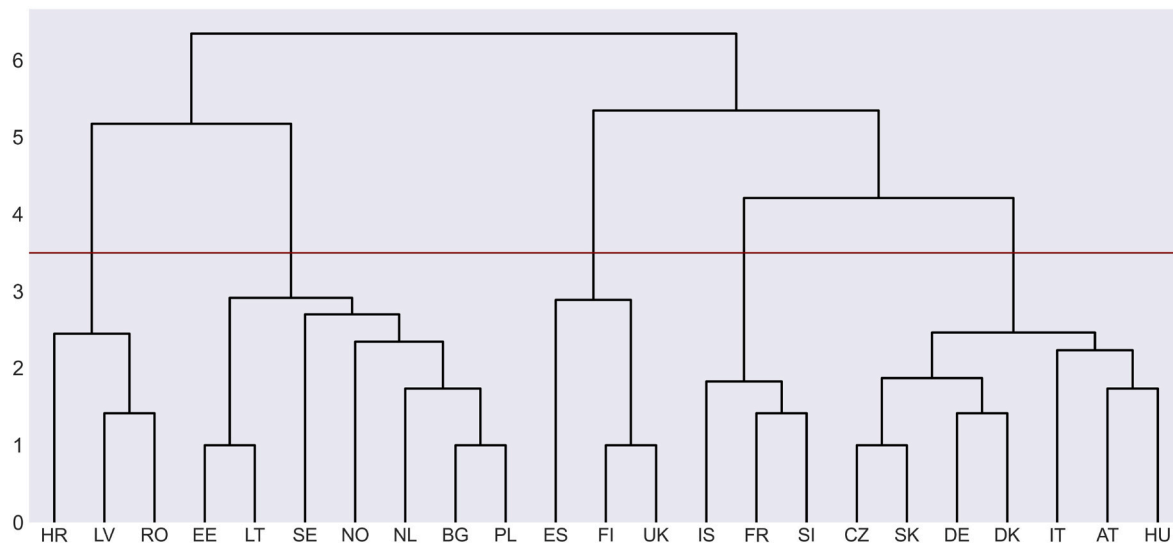


Fig. 8. Hierarchical clustering dendrogram (red line indicates the terminus, i.e. number of clusters). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

initially provided for feasibility studies. In a further module, the implementation of decarbonised DH networks can be subsidised with up to 50% of the expenditures. In addition, customer information measures to increase the connection rate as well as activities of educational and research institutions can be supported through a grant.

In addition, the emissions of heat generation plants with more than 20 MW feeding into DH systems fall under the European Emission Trading System (EU ETS). The EU ETS is regulated by the ETS Directive and also operates in Norway and Iceland, hence, all countries considered in this paper are covered by this scheme. However, some countries work with additional national carbon taxes (World Bank, 2021). Depending on their design, these national carbon taxes can result in even higher costs for fossil fuel heat generation in DH networks. For example, the national carbon tax in Sweden, is imposed on carbon-intensive fuels with the exception of fuels used to generate heat in cogeneration plants that fall under the EU ETS (World Bank, 2021). Nevertheless, the carbon tax applies to any facilities that remain outside of the EU ETS. The carbon tax in Denmark also includes exemptions for operators covered by the EU ETS, however, this exemption does not include DH plants (World Bank, 2021). Another system can be observed in the UK, where a carbon price floor is used in the power sector, including cogeneration feeding into DH grids (World Bank, 2021). This price floor is set to impose a minimum price to the EU ETS market price.

In conclusion, the design of the support systems and carbon taxes vary greatly, which makes it particularly difficult to create a classification that covers all relevant aspects. Nevertheless, the classification developed in this paper (see Fig. 6), builds on the categorization in Bacquet et al. (2022b) and combines it with literature focusing on support for renewable energy in general (e.g. Winkler et al., 2016). The elements of the parameter 'Addressed area of DH' are again used for the overview in section 4.

4. Results and discussion

This section presents an overview of the policy frameworks of DH in the 23 countries and discusses their impact on the share of renewable and waste heat in DH and the share of citizens served by DH.

4.1. Overview of applied policy frameworks

Fig. 7 provides an overview of the policy frameworks, whereby the countries are ordered along their cluster. The corresponding

dendrogram is shown in Fig. 8, which is used to select an appropriate number of clusters. Five clusters, i.e. five distinct policy frameworks, have been selected (see appendix C for further details). Fig. 9 locates the clusters on a map.

Table 1 presents the categories used in Fig. 7 (i.e. values used in the clustering), which are derived from the classification in section 3. The categories (1, 2, 3) represent the elements of the first parameter in the respective morphological box (see also appendix C for further details).

In the framework of the defined categories, Croatia, Latvia, and Romania form the first group (cluster 1) as they follow overall a rather similar policy framework for DH (see Figs. 7 and 9). All three countries apply a high level of regulation but have a low level of support and either no or a low carbon tax (standardized with PPP). The next group (cluster 2) encompasses seven countries: Estonia, Lithuania, Sweden, Norway, the Netherlands, Bulgaria, and Poland. This group regulates TPA with obligations to connect third parties, if specific criteria are fulfilled. At the same time, these countries have more regulation and support in place overall, while using a low level or no carbon tax. Spain, Finland, and the UK form the next group (cluster 3) as they all apply a low level of regulation but rely on a higher level of support and a higher carbon tax. Similarly, Slovenia, France, and Iceland (cluster 4) also introduced a higher carbon tax, but in contrast, these countries rely on stricter regulation, except for TPA. The last group (cluster 5) includes seven countries: Czech Republic, Slovakia, Germany, Denmark, Italy, Austria, and Hungary. They apply rather more regulation, except for TPA, but at the same time, they all implemented a comprehensive support scheme, but do not rely on a carbon tax (except for Denmark).

In addition to the clusters, Fig. 7 visualises which form of regulations and support schemes are most commonly used in the 23 countries. Regarding regulation of ownership, the majority, i.e. 16 out of 23 countries rely on an authorization obligation (category 3). Regarding price regulation 11 out of 23 countries apply rather strict regulation (category 3), where the price is set by the regulator or defined by the operator with ex-ante price control. The most common policy for metering is standards (category 2). For consumer grid connection, all categories are used almost equally. For TPA, 12 out of 23 have not (yet) implemented regulations (category 1). However, in nine countries, TPA is regulated with an obligation to enable grid access for all heat generators (category 3). Regarding support, 16 out of 23 provide comprehensive support (category 3). Finally, 12 countries use a national carbon tax (categories 2 and 3).

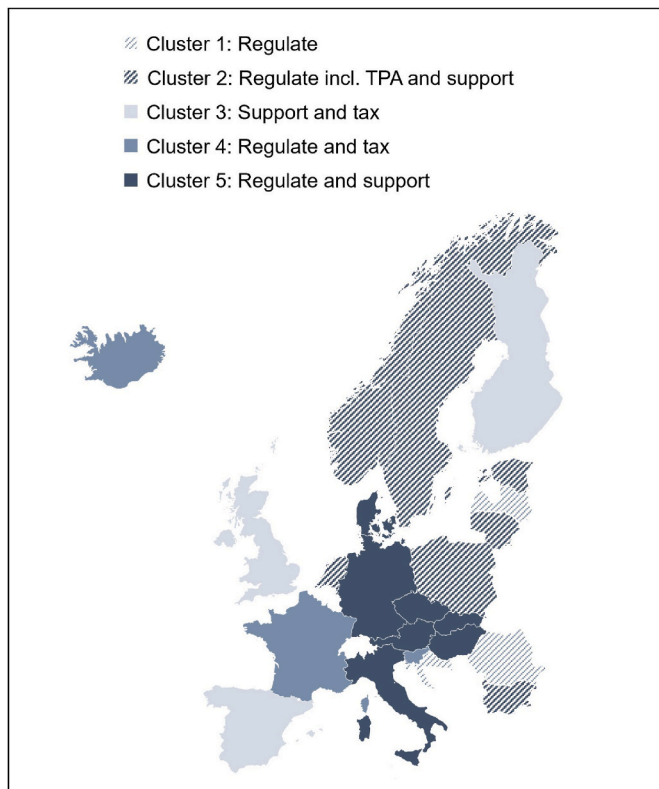


Fig. 9. Map of the clusters (i.e. the five distinguishable policy frameworks).

4.2. Impact of policy frameworks

In order to assess the impact of the policy frameworks, we evaluate the countries in the context of two indicators, i.e. share of renewable and waste heat in DH (including heat pumps) and share of citizens served by DH (Figs. 10 and 11, appendix A). Leaders in the use of DH are Iceland, Denmark, Lithuania, Estonia, Finland, Sweden, and Poland. Countries with remarkably high shares of more than 40% of renewable and waste heat (including heat pumps) are Iceland, Lithuania, Denmark, Sweden, France, Norway, Austria, Latvia, and Estonia.

As presented in Figs. 10 and 11, the share of renewable and waste heat and the share of citizens served are very heterogenic within the clusters. In particular, Fig. 11 shows that no clear correlation can be observed. For example, in cluster 1, the share of renewable and waste heat in DH is only 1% in Romania and Croatia, but 47% in Latvia, even though all three countries follow a similar, more regulated approach. On average, cluster 4, i.e. Slovenia, Iceland, and France, which has more regulation and support in place, shows the highest share of renewable and waste heat in DH. However, this is due to the fact that cluster 4 includes Iceland, which has an 88% share of renewables due to its high geothermal potential. Cluster 2 and 5, comprising countries with more regulation and support, show on average a higher share of renewable and waste energies in DH compared to the other clusters (except cluster 4). Especially cluster 2 shows a high performance with three countries in the upper field, i.e. shares above 40% (compare Fig. 11). These results may indicate that policy frameworks with more regulation and especially with comprehensive support schemes tend to be linked to a higher share of renewable and waste heat in DH. Cluster 2 and 5 also have on average a higher share of citizens served. Thereby, several of the countries in cluster 2 and 5 rely on mandatory grid connection for consumers. These results show that there might be a tendency for stricter regulation and especially obligations for consumer connection to be linked with high connection rates. However, we could also conclude that countries with larger DH markets may have so far introduced more

Table 1

Categories used for the overview, i.e. clustering of the different policies.

Issue	Perceived categories (i.e. elements of the morphological boxes)
Ownership	(1) No specific regulation, (2) Registering, (3) Authorization obligation for networks
Prices	(1) No specific regulation (only cartel law) (2) Price is defined by DH operator with a price calculation rule (3) Price is defined/set by the regulator or defined by the operator with ex-ante price control/approval of the price
Metering	(1) No regulation, (2) Regulation, i.e. mandatory/standards, (3) Specific regulations for smart heat metering and remote control
Consumer grid connection	(1) No regulation, (2) Partial regulation with voluntary connection but for example some regulation of the connection process, (3) Mandatory connection under certain conditions (e.g. zoning)
Third party access (TPA)	(1) No regulation, (2) Regulated TPA with an obligation to enable grid access for renewable or waste heat producers, (3) Regulated TPA with an obligation to enable grid access for all heat generators
Support for DH	(1) No support available, (2) Support for renewables and efficient generation in DH, (3) Support for renewable and efficient generation in DH as well as for DH infrastructure
National carbon tax	(1) No national carbon tax, (2) Carbon tax below 20 \$/t CO ₂ (in 2018; standardized with Purchasing Power Parity (PPP)) (3) Carbon tax higher 20 \$/t CO ₂ (in 2018; standardized with PPP)

legislation. Above all, the differences to the other clusters are small and no clear connection between policy framework and share of renewable and waste heat or share of citizens served can be proven.

The three countries with the highest share of renewable and waste heat in DH and the highest share of citizens served are spread across the clusters: Iceland in cluster 4, Lithuania in cluster 2 and Denmark in cluster 5. These countries generally take a rather regulated approach towards DH and rely on some form of support scheme. However, only Lithuania introduced TPA and only Iceland and Denmark work with a carbon tax, i.e. there are some differences in their policy frameworks. This is an indication that policies are not the only thing that is important for successful DH markets. Among other things, historical development and past policies, population density, gas dependency, the role of combined heat and power or nuclear, involvement of local authorities, consumer perception, heating degree-days, or the availability of renewable potentials may play an important role.

To emphasise that other factors are relevant for successful DH markets, we provide examples from different countries. Estonia (cluster 2) applies a lot of legislation within their national District Heating Act, which already came into force in July 2003. Meanwhile, Estonia reaches a share of renewable and waste heat in DH of 46% and a share of citizens served by DH of 54%, which is above the average of all the countries. Poland (cluster 2) also follows a more regulated approach similar to Estonia and reaches a high share of citizens served by DH. In both countries, this might be reasoned in past policies, i.e. socialism was an important driver for DH. However, with only 9% Poland has (so far) no success in high shares of renewable and waste heat. The success in Estonia could be due to their early focus on efficient and renewable cogeneration. Feed-in tariffs were already introduced in 2007 to promote combined heat and power production. Furthermore, Poland is known for its high coal dependency, whereas Estonia has a larger forestry industry, which contributes significantly to the high share of biomass in DH in Estonia.

On the other side of the spectrum, Finland (cluster 3) follows a less regulated approach compared to Estonia and Poland, but also reaches a high share of 36% of renewable and waste heat in DH and 51% of citizens served. However, the UK (cluster 3) also relies on a less regulated approach, but only reaches a share of renewable and waste heat in DH of around 12%. Thus, both countries rely on a policy framework with few regulations, but the share of renewable and waste energies is quite

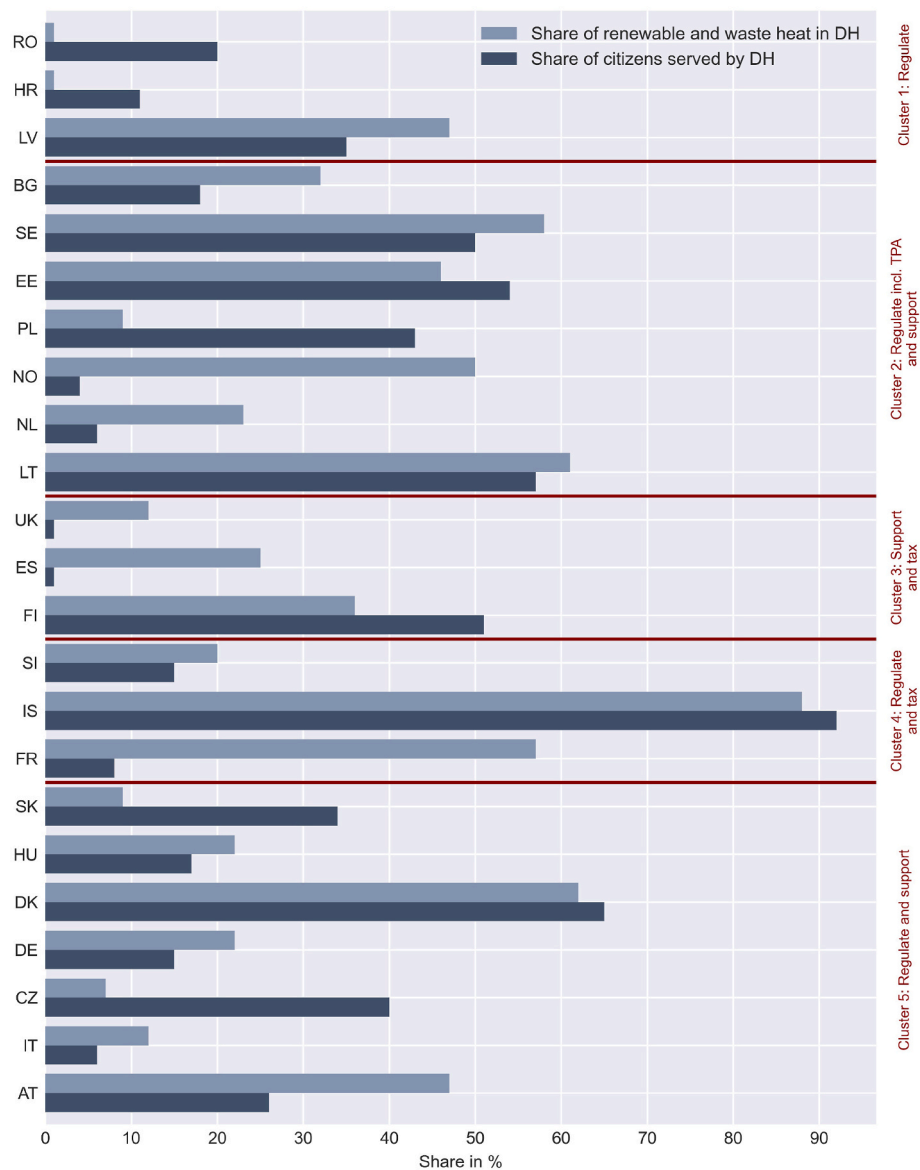


Fig. 10. Share of citizens served by DH and share of renewable and waste heat in DH in 2018 (ES based on 2016) (Bacquet et al., 2022b; Euroheat & Power, 2019; Fleiter et al., 2016).

different. The shortage of regulation in the UK could be due to their rather small DH market, while the trend of urbanisation in Finland might explain the substantial growth of DH in recent years. Another example is Sweden (cluster 2), which introduced TPA, but overall, follows a more market-orientated approach towards DH. At the same time, Sweden reaches a share of 58% renewable and waste heat in DH and 50% of citizens served. Sweden had a very early focus on climate policies, already introducing a carbon tax in 1991. Sweden also has a large forest industry and a lower dependency on oil compared to neighbouring countries, which explain the successful DH market in Sweden.

In summary, the analysis shows that in the majority of the countries the regulation currently in force does not coincide with high shares of DH and high shares of renewable and waste heat in DH (compare Figs. 10 and 11). Looking at countries with higher shares, the analysis shows that a high degree of regulation as well as less regulatory intervention in DH markets can coincide with high or low shares of citizens served by DH and shares of renewable and waste heat in DH. Thus, it is important to recognize that other factors also have a significant impact on the success of DH markets. For example timing and design of regulations matter as well as natural, economic, social and cultural

conditions (e.g. renewable potentials, gas grid development and markets and perceptions of users etc.).

5. Conclusion and policy implications

This paper describes, compares, and groups policy frameworks for DH in Europe. Thereby, a classification for policy aspects addressing DH is developed, comprising regulation of ownership and operation, prices, metering, consumer grid connection, TPA as well as support schemes and carbon taxes (see section 3). The classification builds on existing literature and is further refined on the basis of empirical evidence from 23 countries, collected with an online survey and interviews. Policy-makers can use the classification to better understand the different forms, intensities and varieties of regulatory and support options and assess their national policy mix addressing DH. This is particularly relevant in view of the fact that the European Commission is promoting efficient DH and it is therefore assumed that European DH markets will grow in the future. The growth will be accompanied by an increasing need for regulation, which addresses amongst other things the challenges related to the natural monopoly characteristics of DH grids or the

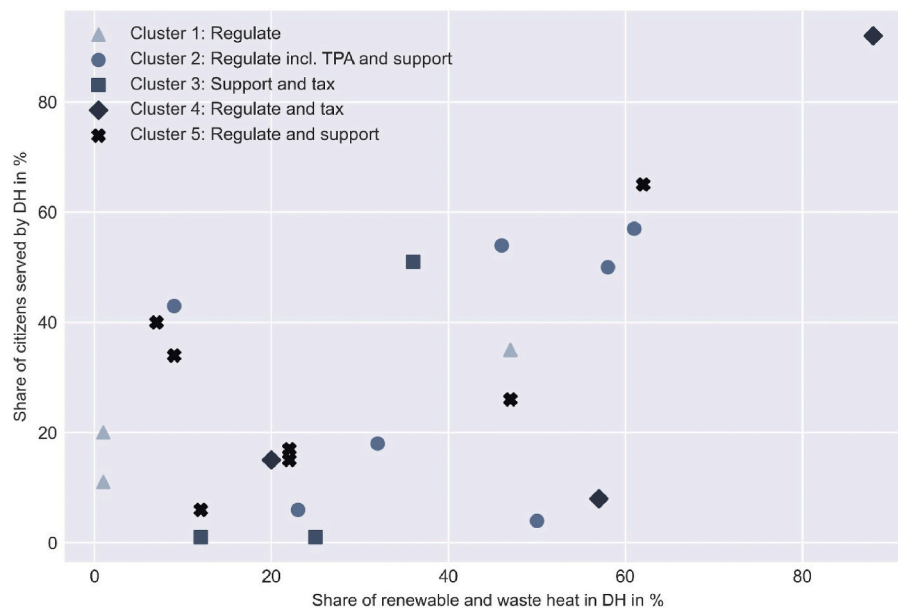


Fig. 11. Comparison of share of citizens served by DH and share of renewable and waste heat in DH in 2018 per cluster (ES based on 2016) (Bacquet et al., 2022b; Euroheat & Power, 2019; Fleiter et al., 2016).

economic viability of efficient and decarbonised DH.

In this paper, the countries are clustered along their assigned categories representing a specific form of regulation or support (see section 4). In this way, five clusters, i.e. five policy frameworks were identified. From a general perspective, these frameworks differ in their intensity of regulation and support and help policymakers to identify their own country's position compared to others. In the following analysis, the five main policy frameworks are embedded in the context of the integration of renewable and waste heat in DH and the diffusion of DH. This analysis concludes that there might be a tendency for policy frameworks with more regulation and support to be linked to a higher share of renewable and waste heat in DH and a higher share of citizens served by DH. However, only a tendency, but no clear correlation could be detected. More data, for example, in terms of historical development of policies as well as growth rates of shares of renewables and waste heat in DH and shares of citizens served by DH (i.e. policies and indicators as a function of time), would be needed for clear conclusions to be drawn, but such data is currently lacking.

Our main conclusion is that the policy frameworks for DH differ strongly in their regulation depth. On the one hand, there are countries with a high degree of regulation (regarding prices, TPA, etc.), on the other hand, there are countries with less regulatory intervention in DH markets. However, neither the share of citizens served by DH nor the share of renewables and waste heat in DH depends on the regulation depth. High shares can be observed in both cases, countries with much and with little DH regulation. In addition to regulation, other facts like high renewable potentials, suitable climatic conditions, low dependency on oil and gas or in general an early focus on climate policy could also be the reason for successful DH markets in some countries. A key policy recommendation is therefore that the country's current situation in DH and in general must be considered when designing suitable policies for DH.

The overview and analysis of the policy frameworks allow for several conclusions and implications for DH policies to be drawn, which can be of great use for policymakers. In several countries, DH networks require some kind of authorisation, with the process and scope of the authorisation and the timing varying depending on the form of ownership (i.e. private or public) and the size of the network. In terms of consumer prices, more than half of the countries have consistent price rules or rely on price control mechanisms (i.e. ex-ante or ex-post) that take into

account the natural monopolistic characteristics of grid-bound heat supply and the potential abuse of market power. Thereby, the depth of regulation varies considerably. In combination with ownership, pricing rules might be less strict if the ownership regulations take into account market failure and include, for example, a non-profit rule for the DH operator.

Metering is also regulated in most of the countries, whereby most countries apply the requirements of the European legislation. However, regulation may differ in terms of the type of meter, frequency and treatment of individual measurements. Smart heat metering and remote control are regulated in only a few countries. Further research would greatly benefit from more data and transparency in DH markets, which could be called for in national and European legislation. Regulation of metering and especially smart individual metering is thereby one regulatory aspect that could improve data availability. Furthermore, data accessibility and market transparency could also be improved, for example, by strengthening information requirements for DH companies. The establishment of an overarching agency that collects data and acts as a central contact point could have a positive impact on transparency and at the same time on consumer perception.

The willingness of consumers to connect to networks is of great importance for DH. To achieve economic efficiency, ensure low prices for consumers and a certain return on investment, a minimum number of consumers need to be connected. Thus, some countries introduced mandatory grid connection for consumers under certain conditions. The design of the obligation varies from country to country and can be aimed at different consumers such as new buildings in specific areas or only households and not industry. Above all, a mandatory connection might only be implementable if the network fulfils certain criteria, such as a high share of renewable and waste heat. As mandatory connection exacerbates the problem of market power, policymakers should ensure appropriate pricing rules or forms of ownership to reduce the risk of abuse of market power. Even though mandatory consumer connections coincide in some countries with a high share of consumers being connected, no clear statistical relationship can be shown in the analysis of this paper. As an alternative to mandatory connection, consumer participation could be improved. Greater customer participation in DH systems, for example through ownership structures or financial participation, could address the natural monopoly and economic viability challenge.

Another aspect is the access of independent producers to DH networks (i.e. TPA). Unlike the gas and electricity market, DH markets are not required to be unbundled under European legislation. However, access of third parties to DH networks is currently being discussed more intensively at the EU level in the framework of the RED II revision. Presently, in about half of the countries, TPA is regulated in some form. These regulations differ significantly in depth and design, but generally, they include an obligation to connect third party heat generators if specific requirements are met, e.g. if this is technically and economically feasible for the respective network. TPA might not only extend to the connection and feed-in of heat, but also to the delivery and sale of heat directly to the end consumers. Such an expansion of TPA may relativize the need for strict pricing rules and authorizations, as it opens the market to new players and thus breaks the monopoly. Yet, regulation on TPA is not well developed in most countries. The analysis of this paper has not shown a clear relationship between TPA and a high share of renewable and waste heat in DH. Nevertheless, TPA can create new dynamics in DH markets and might facilitate the participation of consumers. Thus, TPA to DH networks should be the subject of further research.

Alongside the regulatory aspects, several countries have implemented support schemes for DH networks based on efficient and renewable generation or carbon taxes, which address economic viability issues of decarbonised DH networks as well as the negative external effects of fossil fuel-based systems. In this context, some countries have introduced comprehensive support schemes encompassing several parts of DH such as generation, infrastructure and consumer connection. The analysis shows that especially countries with a comprehensive support system tend to achieve a high share of renewable and waste heat in DH. Thus, for successful DH markets, we recommend direct financial support for decarbonised DH or sufficiently high (national or European) carbon taxes. Quotas for renewables in DH networks might also be a suitable policy measure. Further policy tools to increase the share of renewable and waste heat in DH as well as the diffusion of DH are obligations for strategic long-term heat planning, i.e. encouraging and supporting municipalities in heat planning, including the further use of DH and possible synergies with different infrastructures. Strategic heat plans should be supplemented with long-term transformation plans for existing DH networks.

As this discussion demonstrates, there are interdependency between the different regulations and support measures. Policies may address challenges related to the natural monopoly characteristics of DH, the economic viability of efficient and decarbonised DH or the external effects of fossil fuel based generation. To ensure a functioning and widely accepted DH market, different regulatory aspects and support measures can be combined and their design can influence their required intensity. For example, the need for strict price rules depends on the design of the ownership regulations, consumer connection and TPA, while the extension of TPA may be linked to the form of ownership. A key

conclusion is therefore that the combination and design of regulations and support measures is highly relevant and should be coherent.

Further research focusing on favourable conditions for efficient and renewable DH is needed to advance the understanding of the implementation of decarbonised DH networks. In particular, a more in-depth analysis of successful DH markets (e.g. in Denmark, Lithuania or Estonia), which includes a historical dimension of policies and indicators as well as further regulations, regarding for example spatial planning, could make an additional decisive contribution, building on the work of this paper.

CRediT authorship contribution statement

Anna Billerbeck: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Data curation, Writing – original draft. **Barbara Breitschopf:** Conceptualization, Supervision, Writing – review & editing. **Jenny Winkler:** Conceptualization, Supervision, Writing – review & editing. **Veit Bürger:** Data curation, Writing – review & editing. **Benjamin Köhler:** Data curation, Writing – review & editing. **Alexandre Bacquet:** Data curation, Writing – review & editing. **Eftim Popovski:** Data curation, Writing – review & editing. **Mostafa Fallahnejad:** Data curation, Writing – review & editing. **Lukas Kranzl:** Writing – review & editing. **Mario Ragwitz:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this paper can be found in the appendix and in [Bacquet et al. \(2022a\)](#) and [Bacquet et al. \(2022b\)](#).

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Abbreviations

Art	Article
AT	Austria
BG	Bulgaria
CZ	Czech Republic
DE	Germany
DH	District heating
DHC	District heating and cooling
DK	Denmark
EE	Estonia
EED	Energy Efficiency Directive
ES	Spain
EU ETS	European Emission Trading System

EU	European Union
FI	Finland
FR	France
HR	Croatia
HU	Hungary
IS	Iceland
IT	Italy
LT	Lithuania
LV	Latvia
MS	Member State
MW	Megawatt
NL	Netherlands
NO	Norway
PL	Poland
PPP	Purchasing Power Parity
RED	Renewable Energy Directive
RO	Romania
SE	Sweden
SI	Slovenia
SK	Slovakia
TPA	Third party access
UK	United Kingdom

Appendix A. Share of renewable and waste heat (including heat pumps) in DH and share of citizens served by DH (i.e. indicators used in Fig. 10) (Bacquet et al., 2022b; Euroheat and Power, 2019; Fleiter et al., 2016)

Country	Cluster	Share of renewable and waste heat in DH	Share of citizens served by DH
RO	1	1%	20%
HR		1%	11%
LV		47%	35%
BG	2	32%	18%
SE		58%	50%
EE		46%	54%
PL		9%	43%
NO		50%	4%
NL		23%	6%
LT		61%	57%
UK	3	12%	1%
ES		25%	1%
FI		36%	51%
SI	4	20%	15%
IS		88%	92%
FR		57%	8%
SK	5	9%	34%
HU		22%	17%
DK		62%	65%
DE		22%	15%
CZ		7%	40%
IT		12%	6%
AT		47%	26%

Appendix B. List of organisations that participated in the survey or an interview

Country	Organisation
Austria	Fachverband der Gas- und Wärmeversorgungsunternehmen
Bulgaria	EVN Bulgaria
Croatia	University of Zagreb, Faculty of Mechanical Engineering and Naval Architecture
Czech Republic	Energy Regulatory Office
	Taures A.s.
Denmark	Rambøll
	PlanEnergi
Estonia	Konkurentsiamet
	Tarbijakaitse ja Tehnilise Järelevalve Amet
Finland	Energiateollisuus
France	Fédération des Services Energie Environnement

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Country	Organisation
Germany	Klimaschutz- und Energieagentur Baden-Württemberg Klimaschutzagentur Mannheim Ministerium für Umwelt, Klima und Energiewirtschaft Baden-Württemberg Senatsverwaltung für Wirtschaft, Energie und Betriebe Berlin Bundeskartellamt
Hungary	FÖTÁV Távhőszolgáltatási Divízió Szegedi Távfűtő Kft. (SZETÁV) ÉMI Non-profit Ltd.
Italy	Istituto nazionale di statistica Associazione Italiana Riscaldamento Urbano Gestore dei Servizi Energetici
Latvia	Fortum Oyj
Lithuania	Lietuvos šilumos tiekėjų asociacija Valstybinė energetikos reguliavimo taryba
Netherlands	Netherlands Enterprise Agency
Poland	Izba Gospodarcza Ciepłownictwo Polskie
Romania	B Lighting SRL
Slovakia	Non-disclosable
Slovenia	Jožef Stefan Institute
Spain	IDAE - Instituto para la Diversificación y Ahorro de la Energía
Sweden	Halmstad University Energiföretagen Sverige (Swedenergy) Energimyndigheten
United Kingdom	Association for Decentralised Energy
Iceland	Orkustofnun Norðurorka
Norway	Norges vassdrags- og energidirektorat Norsk Fjernvarme Enova SF

Appendix C. Details on the cluster analysis

This appendix provides additional methodological details on the cluster analysis, including a sensitivity analysis based on different clusterings.

(i) Input data

Input data for the clusterings is an ordinal dataset that was developed based on the morphological analysis in this paper (compare section 3). Table A 1 shows the data set.

Table A1
Input data for the clustering

Country	Ownership	Prices	Metering	Consumer grid connection	Third party access	Support for DH	National carbon tax
Austria (AT)	2	2	2	1	1	3	1
Bulgaria (BG)	3	3	2	2	3	3	1
Czech Republic (CZ)	3	2	2	2	2	3	1
Germany (DE)	2	2	2	3	1	3	1
Denmark (DK)	3	2	2	3	1	3	2
Estonia (EE)	3	3	3	3	3	3	2
Spain (ES)	3	1	1	1	1	3	3
Finland (FI)	1	2	2	1	1	2	3
France (FR)	3	3	2	3	1	3	3
Croatia (HR)	3	3	2	1	1	1	1
Hungary (HU)	3	3	2	2	1	3	1
Iceland (IS)	3	3	1	3	1	2	2
Italy (IT)	2	1	1	2	1	3	1
Lithuania (LT)	3	3	3	3	3	3	1
Latvia (LV)	2	3	2	1	3	1	2
Netherlands (NL)	3	2	2	1	3	3	1
Norway (NO)	3	2	1	3	3	3	2
Poland (PL)	3	3	2	2	3	3	2
Romania (RO)	3	3	2	1	3	1	1
Sweden (SE)	1	2	2	2	3	3	2
Slovenia (SI)	3	3	2	2	1	2	3
Slovakia (SK)	3	2	2	3	2	3	1
United Kingdom (UK)	1	1	2	1	1	2	3

The values (1, 2, 3) represent different categories, which are explained in Table 1 in section 4. For example, for the variable "Ownership", value one refers to "No specific regulation of ownership"; value two refers to "Requirement for registration for operators" and value three refers to "Authorization, i.e. license, permit or concession". These categories are the elements of the parameter "Main form of regulation" in the morphological analysis for the regulatory issue "Ownership and operation of DH" (compare Fig. 1 in section 3.1).

The categories are thereby different for each regulatory issue (compare Table 1 in section 4). However, for all regulatory issues, the categories represent an increased policy intensity, i.e. value one refers to a low policy intensity and value three refers to a strong policy intensity, which leads to an ordinal data set.

In clustering, ordinal data is often coded as consecutive integers and treated as numerical ones, because of a lack of established distance metrics for ordinal data (see e.g. Ranalli, Monia and Rocci, Roberto, 2015; Stein and Vollnhals, 2011). In line with this common approach, we treat the ordinal data (Table A 1) as numerical by assuming identical distances between the categories.

(ii) Clustering

Clustering or cluster analyses is a collective term for statistical procedures that make it possible to structure a data set by group assignments. Cluster analyses are based on the consideration of the similarity or distance of the objects to each other. There are a variety of different algorithms that use different distance measures to assign objects to clusters. In general, partitioning (including k-means), hierarchical, density-based, grid-based and combined methods are distinguished (compare e.g. Stein and Vollnhals, 2011).

In this paper, we performed hierarchical clustering as well as k-means clustering. The hierarchical clustering was performed using algorithms provided by the Python package SciPy. Further information, including formulas for the linkage types and dissimilarity calculations, can be found in the documentation of SciPy.⁵ The k-means clustering was computed using the k-means++ algorithm provided by the Python package Scikit-learn. Further information is available in the documentation and user guide.⁶

For the hierarchical clustering algorithms, we use the cophenetic correlation coefficient⁷ to compare different linkage types and dissimilarity calculations (see Table A2). This coefficient calculates the cophenetic distances between each observation in the hierarchical clustering defined by the linkage. It measures the height of the dendrogram at the point where two branches merge. The magnitude of this value should be close to one for a high-quality solution.

Table A2

Cophenetic correlation coefficient for different hierarchical clusterings

metric \ method	ward ⁸	single	complete	average	weighted
euclidean	0.627	0.663	0.535	0.766	0.727
cityblock (Manhattan distance)	-	0.626	0.584	0.747	0.646
hamming	-	0.467	0.533	0.547	0.526
correlation	-	0.446	0.633	0.654	0.640
mahalanobis	-	0.553	0.486	0.561	0.555
chebyshev	-	0.589	0.471	0.677	0.651

For further analysis, we choose per linkage method up to two metrics with the highest cophenetic correlation coefficient. For the sensitivity analysis, we also computed k-means clustering. Thus, the following sensitivities are computed:

- S0: Hierarchical clustering with Ward's minimum variance method and Euclidean distance
- S1: Hierarchical clustering with single linkage and Euclidean distance
- S2: Hierarchical clustering with single linkage and Manhattan distance
- S3: Hierarchical clustering with complete linkage and correlation distance
- S4: Hierarchical clustering with average linkage and Euclidean distance
- S5: Hierarchical clustering with average linkage and Manhattan distance
- S6: Hierarchical clustering with weighted linkage and Euclidean distance
- S7: K-means clustering

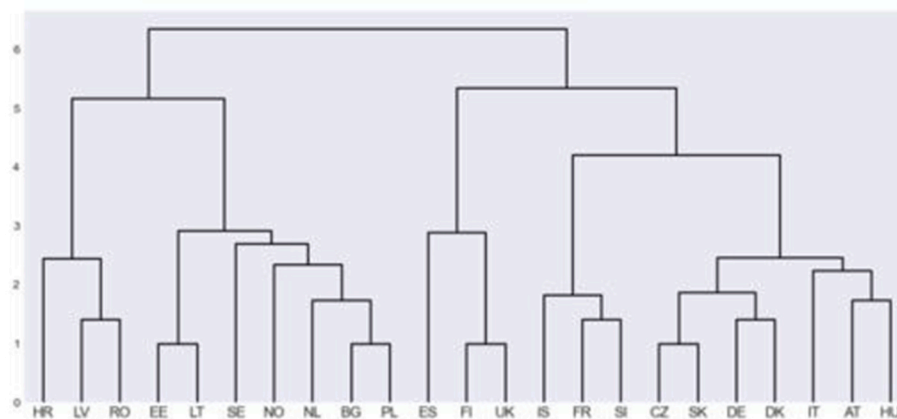
In hierarchical clustering, the terminus, i.e. the number of clusters, can be chosen at will (compare e.g. Stein and Vollnhals, 2011). So-called dendrograms, which show the pairwise combination of elements over dissimilarity, are used to select a terminus, i.e. the number of clusters. To make the selection of the number of clusters and the resulting division of the countries transparent, figure A 1 shows the dendrograms of the hierarchical clusterings.

⁵ See <https://docs.scipy.org/doc/scipy/reference/generated/scipy.cluster.hierarchy.linkage.html#scipy.cluster.hierarchy.linkage> and <https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.pdist.html#scipy.spatial.distance.pdist>.

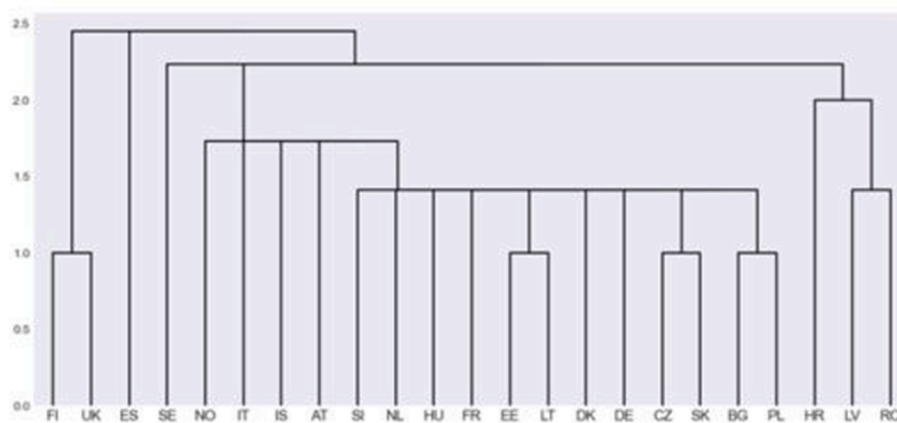
⁶ See <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html> and <https://scikit-learn.org/stable/modules/clustering.html#k-means>.

⁷ Based on the package SciPy; <https://docs.scipy.org/doc/scipy/reference/generated/scipy.cluster.hierarchy.cophenet.html>.

S0: Hierarchical clustering with Ward's minimum variance method and the Euclidean distance



S1: Hierarchical clustering with single linkage and Euclidean distance



S2: Hierarchical clustering with single linkage and Manhattan distance

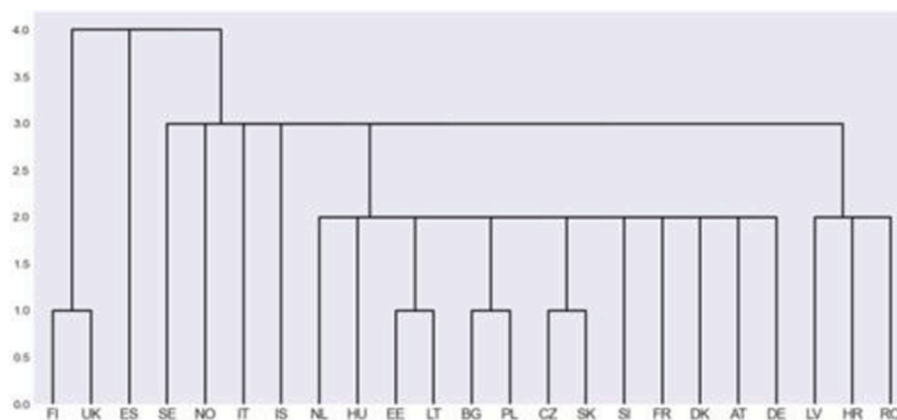
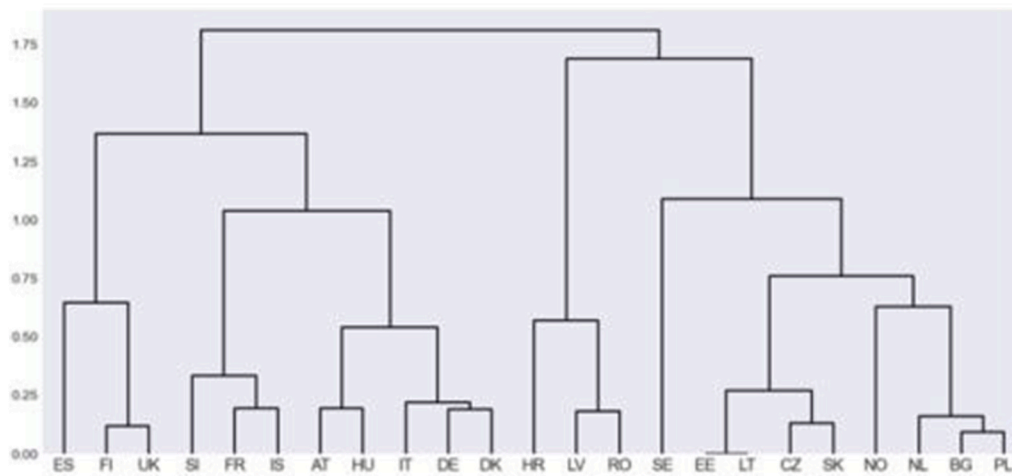
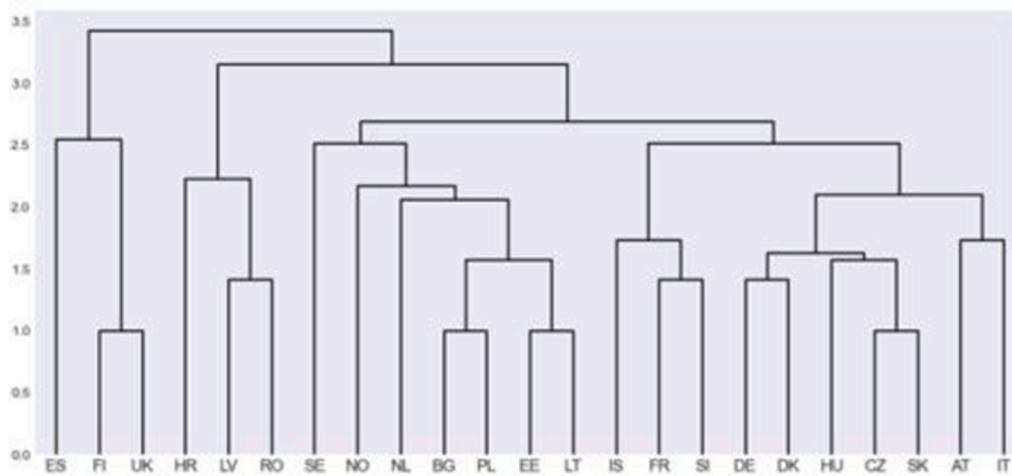


Fig. A1. Dendrograms of the different hierarchical clusterings

S3: Hierarchical clustering with complete linkage and correlation distance



S4: Hierarchical clustering with average linkage and Euclidean distance



S5: Hierarchical clustering with average linkage and Manhattan distance

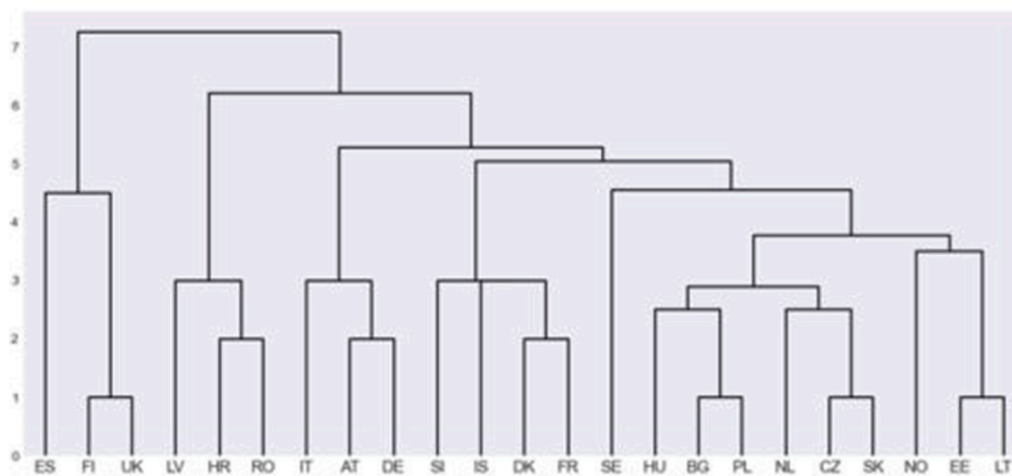


Fig. A1. (continued).

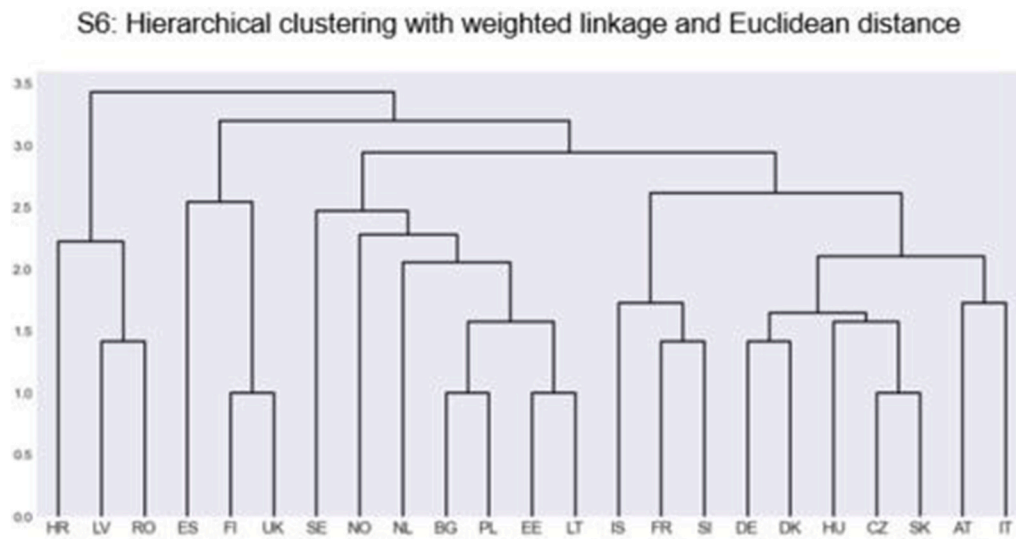


Fig. A1. (continued).

In the following, we compare the results of different termini, i.e. a different number of clusters (three, four and five clusters). Table A 3 shows the results, i.e. the resulting country cluster per clustering, by choosing three clusters. With three clusters, S4, S5 and S6 have the same results. In addition, S1 and S2 have identical results. Moreover, the majority of clusterings group HR, LV, RO into one cluster and ES, FI and UK into another cluster (as part of a bigger cluster).

Table A3
Country groups by choosing three clusters

Cluster	S0	S1	S2	S3	S4	S5	S6	S7
1	HR, LV, RO, EE, LT, SE, NO, NL, BG, PL	FI, UK	FI, UK	ES, FI, UK, SI, FR, IS, AT, HU, IT, DE, DK	ES, FI, UK	ES, FI, UK	HR, LV, RO	AT, DE, DK, ES, FI, FR, HU, IS, IT, SI, UK
2	ES, FI, UK	ES	ES	HR, LV, RO	HR, LV, RO	LV, HR, RO	ES, FI, UK	BG, CZ, EE, LT, NL, NO, PL, SE, SK HR, LV, RO
3	IS, FR, SI, CZ, SK, DE, DK, IT, AT, HU	SE, NO, IT, IS, AT, SI, NL, HU, FR, EE, LT, DK, DE, CZ, SK, BG, PL, HR, LV, RO	SE, NO, IT, IS, AT, SI, NL, HU, FR, EE, LT, DK, DE, CZ, SK, BG, PL, HR, LV, RO	SE, EE, LT, CZ, SK, NO, NL, BG, PL	SE, NO, NL, BG, PL, EE, LT, IS, FR, SI, DE, DK, HU, CZ, SK, AT, IT	IT, AT, DE, SI, IS, DK, FR, SE, HU, BG, PL, NL, CZ, SK, NO, EE, LT	SE, NO, NL, BG, PL, EE, LT, IS, FR, SI, DE, DK, HU, CZ, SK, AT, IT	

Table A 4 shows the country cluster with four clusters. Thereby, S1 and S2 are not included, because for both no terminus with four clusters exists (compare dendrograms in figure A 1). All other clusterings cluster HR, LV, RO into one cluster and ES, FI and UK into another cluster (as part of a bigger cluster). Further, S0, S4 and S6 lead to the same results. Finally, the majority of clusterings cluster EE, SE, LT, NO, NL, BG, PL into one cluster and IS, FR, SI, DE, DK, IT, AT, HU into another cluster (as part of a bigger cluster).

Table A4
Country groups by choosing four clusters

Cluster	S0	S1	S2	S3	S4	S5	S6	S7
1	HR, LV, RO	–	–	ES, FI, UK	ES, FI, UK	ES, FI, UK	HR, LV, RO	DE, DK, FR, HU, IS, SI
2	EE, LT, SE, NO, NL, BG, PL	–	–	SI, FR, IS, AT, HU, IT, DE, DK	HR, LV, RO	LV, HR, RO	ES, FI, UK	BG, CZ, EE, LT, NL, NO, PL, SE, SK
3	ES, FI, UK	–	–	HR, LV, RO	SE, NO, NL, BG, PL, EE, LT	IT, AT, DE	SE, NO, NL, BG, PL, EE, LT	HR, LV, RO
4	IS, FR, SI, CZ, SK, DE, DK, IT, AT, HU	–	–	SE, EE, LT, CZ, SK, NO, NL, BG, PL	IS, FR, SI, DE, DK, HU, CZ, SK, AT, IT	SI, IS, DK, FR, SE, HU, BG, PL, NL, CZ, SK, NO, EE, LT	IS, FR, SI, DE, DK, HU, CZ, SK, AT, IT	AT, ES, FI, IT, UK

Finally, Table A 5 shows the country groups with five clusters. Again, S2 is not included, because no terminus resulting in five clusters exists (compare dendrogram in figure A 1). Choosing five clusters, S0 and S6 have the same results. S5 is quite similar to S0 and S6, however, it computes different clusters for DK, HU and CZ. In contrast, S1, S3 and S4 compute clusters with only one country.

Table A5

Country groups by choosing five clusters

Cluster	S0	S1	S2	S3	S4	S5	S6	S7
1	HR, LV, RO	FI, UK	–	ES, FI, UK	FI, UK	ES, FI, UK	HR, LV, RO	DE, DK, FR, HU, IS, SI
2	EE LT, SE, NO, NL, BG, PL	ES	–	SI, FR, IS, AT, HU, IT, DE, DK	ES	LV, HR, RO	ES, FI, UK	BG, CZ, EE, LT, NL, NO, PL, SE, SK
3	ES, FI, UK	SE	–	HR, LV, RO	HR, LV, RO	IT, AT, DE	SE, NO, NL, BG, PL, EE, LT	HR, LV, RO
4	IS, FR, SI	NO, IT, AT, SI, NL, HU, FR, EE, LT, DK, DE, CZ, SK, BG, PL, HR, LV, RO	–	SE	SE, NO, NL, BG, PL, EE, LT	SI, IS, DK, FR	IS, FR, SI	AT, ES, IT
5	CZ, SK, DE, DK, IT, AT, HU	HR, LV, RO	–	EE, LT, CZ, SK, NO, NL, BG, PL	IS, FR, SI, DE, DK, HU, CZ, SK, AT, IT	SE, HU, BG, PL, NL, CZ, SK, NO, EE, LT	DE, DK, HU, CZ, SK, AT, IT	FI, UK

The comparison of the clustering results with different termini shows that the algorithms can lead to similar results. Despite different clusterings and terminus HR, LV, RO are often clustered into one cluster and ES, FI and UK are clustered into another cluster. In addition, EE, LT, SE, NO, NL, BG, PL are often clustered into one cluster and IS, FR, SI, DE, DK, IT, AT, HU are often clustered into another cluster. This indicates the robustness of the results. In particular, with four clusters, S0, S4 and S6 compute the same clusters, i.e. have the same results. With five clusters, S0 and S6 still compute the same clusters.

Since the overall aim of the paper is to identify different policy approaches, a higher number of clusters is preferred as it represents a greater variety of policy frameworks. Against this background, it seems reasonable to choose four or five clusters for the analysis in this paper. As S0, S4 and S6 have identical results for four clusters and S0 and S6 have identical results for five clusters, it seems reasonable to choose one of these clusterings.

For the final decision, we compare the S0 and S4 clusterings with the Calinski-Harabasz⁸ and the Silhouette score.⁹ We exclude S6 as it is equal to S0 and/or S4 with the respective terminus. The Calinski-Harabasz score is a variance ratio measurement, which measures the ratio between within-cluster dispersion and between-cluster dispersion. A higher Calinski-Harabasz score relates to a model with better-defined clusters. The silhouette coefficient measures how well an observation is clustered and it estimates the average distance between clusters. The best value is one and the worst value is minus one. Values near zero indicate overlapping clusters. The comparison of the two scores shows that S0 with five clusters reaches the highest numbers (compare table A 6).

Table A6

Calinski-Harabasz and the Silhouette score for S0 and S4

Measurement		S0	S4
4 cluster	Calinski-Harabasz	6.939	6.939
	Silhouette	0.232	0.232
5 cluster	Calinski-Harabasz	7.331	5.927
	Silhouette	0.247	0.231

Concluding, S0 with five clusters has identical results to S6, cross-validating this clustering. In addition, it reaches a high cophenetic correlation coefficient and the highest Calinski-Harabasz score and the highest Silhouette coefficient. Against this background, we choose to use the S0 clustering (hierarchical clustering with Ward's minimum variance method and Euclidean distance) with five clusters in the paper.

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⁸ Based on the package SciKit; https://scikit-learn.org/stable/modules/generated/sklearn.metrics.calinski_harabasz_score.html.

⁹ Based on the package SciKit; https://scikit-learn.org/stable/modules/generated/sklearn.metrics.silhouette_score.html.

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